EXTENSIVE MATHEMATICAL DERIVATIONS

Brownian Bridge Diffusion Model

Posterior Derivation and Noise Parameterization

Step-by-step proofs, probability identities,
Gaussian calculations, and mathematical interpretation

PART 2 — BAYESIAN POSTERIOR AND REPARAMETERIZATION

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1 Objective and notation

We derive the exact posterior

$$q_{BB}(x_{t-1} \mid x_t, x_0, y)$$

from Bayes' theorem and the Markov transition established in Part 1. We then derive its variance and mean by multiplying two Gaussian densities and completing the square. Throughout, all distributions are conditional on the fixed endpoints x_0, y, even where the transition notation suppresses them.

Set

$$r_t = \frac{1 - m_t}{1 - m_{t-1}}, \qquad b_t = m_t - r_t m_{t-1}, \qquad v_t = \delta_{t|t-1} = \delta_t - r_t^2 \delta_{t-1}.$$

The transition is

$$q_{BB}(x_t \mid x_{t-1}, x_0, y) = q_{BB}(x_t \mid x_{t-1}, y) = \mathcal{N}(x_t; r_t x_{t-1} + b_t y, v_t I). \tag{1.1}$$

The equality is the Markov property: once x_{t-1} and the endpoint condition y are known, the earlier source endpoint x_0 does not need to be listed separately in the one-step transition. It does not mean x_0 is absent from the overall bridge construction.

The one-time prior is

$$q_{BB}(x_{t-1} \mid x_0, y) = \mathcal{N}(x_{t-1}; \mu_0, \delta_{t-1} I), \quad \mu_0 = (1 - m_{t-1})x_0 + m_{t-1} y. \tag{1.2}$$

2 Bayes' theorem with explicit conditioning

Bayes' theorem gives

$$\boxed{q_{BB}(x_{t-1} \mid x_t, x_0, y) = \frac{q_{BB}(x_t \mid x_{t-1}, x_0, y)\, q_{BB}(x_{t-1} \mid x_0, y)}{q_{BB}(x_t \mid x_0, y)}.} \tag{2.1}$$

The first factor may be shortened by the Markov property:

$$q_{BB}(x_t \mid x_{t-1}, x_0, y) = q_{BB}(x_t \mid x_{t-1}, y). \tag{2.2}$$

This is the correct reason x_0 is omitted in the first factor. The Markov property removes dependence on the earlier history once the current state is known. It does not remove x_0 from the prior $q_{BB}(x_{t-1} \mid x_0, y)$, nor from the posterior's conditioning set.

Since the denominator in (2.1) is independent of the variable x_{t-1}, it is a normalization constant when deriving the posterior density:

$$q_{BB}(x_{t-1} \mid x_t, x_0, y) \propto q_{BB}(x_t \mid x_{t-1}, y) q_{BB}(x_{t-1} \mid x_0, y). \tag{2.3}$$

3 Multiplying the two Gaussian factors

For compactness, write $z = x_{t-1}$. The likelihood factor is

$$q_{BB}(x_t \mid z, y) \propto \exp\left[-\frac{1}{2v_t}\|x_t - r_t z - b_t y\|^2\right], \tag{3.1}$$

and the prior factor is

$$q_{BB}(z \mid x_0, y) \propto \exp\left[-\frac{1}{2\delta_{t-1}}\|z - \mu_0\|^2\right]. \tag{3.2}$$

Their product is

$$q_{BB}(z \mid x_t, x_0, y) \propto \exp\left\{-\frac{1}{2}\left[\frac{\|x_t - r_t z - b_t y\|^2}{v_t} + \frac{\|z - \mu_0\|^2}{\delta_{t-1}}\right]\right\}. \tag{3.3}$$

3.1 Expand both quadratic forms

First,

$$\begin{aligned}\|x_t - r_t z - b_t y\|^2 &= \|r_t z - (x_t - b_t y)\|^2 \\ &= r_t^2 z^\top z - 2r_t z^\top (x_t - b_t y) + \|x_t - b_t y\|^2.\end{aligned} \tag{3.4}$$

Second,

$$\|z - \mu_0\|^2 = z^\top z - 2z^\top \mu_0 + \mu_0^\top \mu_0. \tag{3.5}$$

Substitute (3.4)–(3.5) into (3.3) and collect terms depending on z:

$$q_{BB}(z \mid x_t, x_0, y) \propto \exp\left\{-\frac{1}{2}\left[A_t z^\top z - 2z^\top B_t + C_t\right]\right\}, \tag{3.6}$$

where

$$A_t = \frac{r_t^2}{v_t} + \frac{1}{\delta_{t-1}}, \qquad B_t = \frac{r_t(x_t - b_t y)}{v_t} + \frac{\mu_0}{\delta_{t-1}}, \tag{3.7}$$

and C_t contains all terms independent of z.

4 Completing the square

The scalar-precision identity is

$$A z^\top z - 2z^\top B = A\left\|z - \frac{B}{A}\right\|^2 - \frac{\|B\|^2}{A}. \tag{4.1}$$

To verify it, expand the squared norm:

$$\begin{aligned}A\left\|z - \frac{B}{A}\right\|^2 &= A\left(z^\top z - \frac{2z^\top B}{A} + \frac{B^\top B}{A^2}\right) \\ &= A z^\top z - 2z^\top B + \frac{\|B\|^2}{A}.\end{aligned}$$

Rearranging gives (4.1).

The final term $-\|B\|^2/A$ does not depend on z, so it is absorbed into the normalizing constant. Comparing (3.6) with a Gaussian density gives

$$\boxed{\widetilde{\delta}_t = \frac{1}{A_t}, \qquad \widetilde{\mu}_t = \frac{B_t}{A_t}.} \tag{4.2}$$

Therefore,

$$\boxed{q_{BB}(x_{t-1} \mid x_t, x_0, y) = \mathcal{N}(x_{t-1}; \widetilde{\mu}_t, \widetilde{\delta}_t I).} \tag{4.3}$$

5 Deriving the posterior variance

From (3.7),

$$\widetilde{\delta}_t = \left(\frac{r_t^2}{v_t} + \frac{1}{\delta_{t-1}} \right)^{-1}.$$

Taking a common denominator,

$$\begin{aligned} \widetilde{\delta}_t &= \left(\frac{r_t^2 \delta_{t-1} + v_t}{v_t \delta_{t-1}} \right)^{-1} \\ &= \frac{v_t \delta_{t-1}}{r_t^2 \delta_{t-1} + v_t}. \end{aligned}$$

But $v_t = \delta_t - r_t^2 \delta_{t-1}$, hence

$$r_t^2 \delta_{t-1} + v_t = \delta_t.$$

Thus

$$\boxed{\widetilde{\delta}_t = \frac{\delta_{t|t-1}\, \delta_{t-1}}{\delta_t}.} \tag{5.1}$$

This is the familiar precision-addition rule for multiplying Gaussian densities: precisions add, so the posterior variance is the reciprocal of the sum of precisions.

6 Deriving the posterior mean

Using $\widetilde{\mu}_t = \widetilde{\delta}_t B_t$ and (3.7),

$$\widetilde{\mu}_t = \widetilde{\delta}_t \left[\frac{r_t (x_t - b_t y)}{\delta_{t|t-1}} + \frac{\mu_0}{\delta_{t-1}} \right].$$

Substitute (5.1):

$$\boxed{\widetilde{\mu}_t = \frac{\delta_{t-1} r_t}{\delta_t} (x_t - b_t y) + \frac{\delta_{t|t-1}}{\delta_t} \mu_0.} \tag{6.1}$$

Now insert

$$\mu_0 = (1 - m_{t-1}) x_0 + m_{t-1} y$$

and expand:

$$\begin{aligned}\widetilde{\mu}_t = &\frac{\delta_{t-1} r_t}{\delta_t} x_t - \frac{\delta_{t-1} r_t b_t}{\delta_t} y \\ &+ \frac{\delta_{t|t-1}}{\delta_t}(1 - m_{t-1}) x_0 + \frac{\delta_{t|t-1}}{\delta_t} m_{t-1} y.\end{aligned} \tag{6.2}$$

Collecting coefficients,

$$\boxed{\begin{aligned}\widetilde{\mu}_t = &\frac{\delta_{t-1} r_t}{\delta_t} x_t + \frac{\delta_{t|t-1}}{\delta_t}(1 - m_{t-1}) x_0 \\ &+ \left[\frac{\delta_{t|t-1}}{\delta_t} m_{t-1} - \frac{\delta_{t-1} r_t b_t}{\delta_t}\right] y.\end{aligned}} \tag{6.3}$$

Since $b_t = m_t - r_t m_{t-1}$, the coefficient of y becomes

$$\begin{aligned}&\frac{\delta_{t|t-1} m_{t-1} - \delta_{t-1} r_t (m_t - r_t m_{t-1})}{\delta_t} \\ &= \frac{(\delta_{t|t-1} + r_t^2 \delta_{t-1}) m_{t-1} - \delta_{t-1} r_t m_t}{\delta_t} \\ &= m_{t-1} - \frac{\delta_{t-1} r_t m_t}{\delta_t},\end{aligned}$$

because $\delta_{t|t-1} + r_t^2 \delta_{t-1} = \delta_t$. Hence an equivalent compact form is

$$\boxed{\begin{aligned}\widetilde{\mu}_t = &\frac{\delta_{t-1} r_t}{\delta_t} x_t + \frac{\delta_{t|t-1}}{\delta_t}(1 - m_{t-1}) x_0 \\ &+ \left(m_{t-1} - \frac{\delta_{t-1} r_t m_t}{\delta_t}\right) y.\end{aligned}} \tag{6.4}$$

Equations (5.1) and (6.4) are derived directly from the transition (1.1) and the one-time prior (1.2). If a printed version of Eq. (12) appears in a different algebraic arrangement, compare coefficients after substituting the definitions of $r_t, b_t, \delta_{t|t-1}$; do not assume two visually different forms are unequal.

7 Why the posterior is not directly usable at inference

The exact posterior depends on x_t, x_0, y. During training, x_0 is known, so the posterior is available as a target distribution. During generation, the clean source x_0 is unknown. The model therefore needs a parameterization that predicts a quantity available from the noisy state and conditioning endpoint, such as the forward noise or residual.

8 Noise-prediction reparameterization

From the forward equation,

$$x_t = (1 - m_t) x_0 + m_t y + \sqrt{\delta_t}\epsilon. \tag{7.1}$$

Solving for the clean state,

$$\boxed{x_0 = \frac{x_t - m_t y - \sqrt{\delta_t}\epsilon}{1 - m_t},} \qquad m_t \neq 1. \tag{7.2}$$

Substitute (7.2) into the posterior mean (6.1), using $\mu_0 = (1 - m_{t-1})x_0 + m_{t-1}y$:

$$\begin{aligned} \widetilde{\mu}_t = {} & \frac{\delta_{t-1} r_t}{\delta_t}(x_t - b_t y) \\ & + \frac{\delta_{t|t-1}}{\delta_t}\left[\frac{1 - m_{t-1}}{1 - m_t}(x_t - m_t y - \sqrt{\delta_t}\epsilon) + m_{t-1} y\right]. \end{aligned} \tag{7.3}$$

Since

$$r_t = \frac{1 - m_t}{1 - m_{t-1}}, \quad \text{so} \quad \frac{1 - m_{t-1}}{1 - m_t} = \frac{1}{r_t},$$

the coefficient of ϵ in (7.3) is

$$-\frac{\delta_{t|t-1}}{\delta_t}\frac{1 - m_{t-1}}{1 - m_t}\sqrt{\delta_t} = -\frac{\delta_{t|t-1}}{r_t\sqrt{\delta_t}}.$$

All remaining terms depend only on x_t, y, t. Thus

$$\boxed{\widetilde{\mu}_t = A_t(x_t, y) - \frac{\delta_{t|t-1}}{r_t\sqrt{\delta_t}}\,\epsilon,} \tag{7.4}$$

where $A_t(x_t, y)$ is the deterministic coefficient combination obtained by collecting the non-noise terms in (7.3). A network can replace ϵ with $\epsilon_\theta(x_t, y, t)$, making the reverse mean evaluable without access to x_0.

8.1 Residual conventions and sign discipline

A second useful residual is

$$R_t = x_t - x_0 = m_t(y - x_0) + \sqrt{\delta_t}\epsilon. \tag{7.5}$$

This residual is not identical to the standardized noise ϵ. If a paper predicts R_t, the target and the coefficient multiplying the network output must be derived in that residual convention. If it predicts ϵ, use (7.2)–(7.4). The sign of the noise coefficient follows from solving the forward equation for x_0; it should not be changed by intuition. Definitions of coefficients and signs must be checked together.

9 Summary of posterior results

$$\boxed{q_{BB}(x_{t-1} \mid x_t, x_0, y) = \mathcal{N}(\widetilde{\mu}_t, \widetilde{\delta}_t I)}$$

with

$$\boxed{\widetilde{\delta}_t = \frac{\delta_{t|t-1}\delta_{t-1}}{\delta_t}}$$

and

$$\boxed{\widetilde{\mu}_t = \frac{\delta_{t-1} r_t}{\delta_t}(x_t - b_t y) + \frac{\delta_{t|t-1}}{\delta_t}((1 - m_{t-1})x_0 + m_{t-1} y).}$$

These formulas are the analytical posterior target used to train the learned reverse transition.